

Sensitivity via Omitted Variable Bias

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Warning we'll keep repeating: this is *not* “how much confounding is there.”

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Strategy: SOO conditioning on village and gender.

Darfur: estimate, and the worry

Setup: outcome *PeaceIndex* (0–1); treatment *DirectHarm* (0/1).

$$PeaceIndex_i = \alpha DirectHarm_i + \beta^T Village_i + \phi female_i + \epsilon_i.$$

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- 4 How can we make informed guesses about plausible confounding?

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We focus on OVB-based: scale-free in R^2 , handles multiple/non-linear confounders, and plugs straight into a regression workflow.

A short detour: Frisch–Waugh–Lovell

Suppose we run OLS of Y on D and $\mathbf{X}$:

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In words: *multivariate OLS reduces to univariate OLS, after partialling out $\mathbf{X}$,*

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This is the central tool for what follows.

Omitted variable bias: setup

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Whether the “full” regression is wise, or whether its coefficient is the quantity of interest, is a separate question — here we just compute the consequence of omission.

OVB via FWL: $\widehat{\text{bias}} = \hat{\gamma}\hat{\delta}$

Restating FWL,

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$\hat{\delta}$ is the coefficient on D in the auxiliary regression

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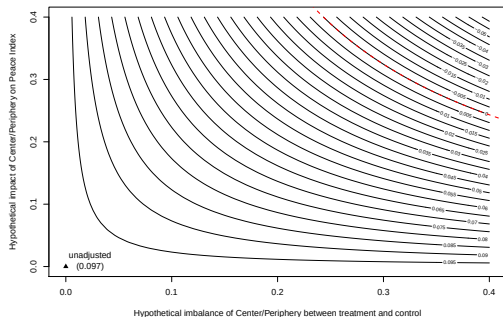
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- Scale-free, bounded in $[0, 1]$
- Lets us derive bias *and* variance, do bounds, and handle all confounders jointly.

What is the partial R^2 ?

The partial correlation is simply correlation *after* partialling out $\mathbf{X}$:

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Direction-symmetric: $R_{D \sim Z | \mathbf{X}}^2 = R_{Z \sim D | \mathbf{X}}^2$.

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Recall the standard OLS variance fact, $\widehat{\text{se}}(\hat{\tau}_{\text{res}}) = \frac{\text{sd}(Y^\perp_{\mathbf{x}, D})}{\text{sd}(D^\perp_{\mathbf{x}}) \sqrt{\text{df}}}$. So equivalently

$$|\widehat{\text{bias}}| = \widehat{\text{se}}(\hat{\tau}_{\text{res}}) \sqrt{\frac{R_{Y \sim Z|D, \mathbf{x}}^2 R_{D \sim Z|\mathbf{x}}^2}{1 - R_{D \sim Z|\mathbf{x}}^2}} \cdot \text{df}.$$

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Everything data-dependent is now in $\widehat{\text{se}}(\hat{\tau}_{\text{res}})$ and df — both already on your regression printout.

Bias anatomy

Consider the *relative bias* — the fraction of $\hat{\tau}_{\text{res}}$ that is bias. Using

$$|\hat{\tau}_{\text{res}}| = |t_D| \widehat{\text{se}}(\hat{\tau}_{\text{res}}),$$

$$\frac{|\widehat{\text{bias}}|}{|\hat{\tau}_{\text{res}}|} = \frac{\sqrt{\text{df}} \cdot \text{BF}}{|t_D|} = \frac{\text{BF}}{|f_{Y \sim D|X}|},$$

where $\text{BF} := \sqrt{\frac{R_{Y \sim Z|D,X}^2 R_{D \sim Z|X}^2}{1 - R_{D \sim Z|X}^2}}$ is the “bias factor” and

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$f_{Y \sim D|X} := R_{Y \sim D|X} / \sqrt{1 - R_{Y \sim D|X}^2}$ is the partial Cohen’s f . Three things to take away:

- BF is the expression of confounder strength.
- $f_{Y \sim D|X}$ — essentially how much of Y ’s variance the treatment uniquely explains — is *all* that enters from the data. It is what confounding is competing to explain.
- $f_{Y \sim D|X}$ is *not* a function of sample size: more data does not help you fight confounding.

Variance anatomy

Applying the same OLS variance fact to the *full* regression and combining:

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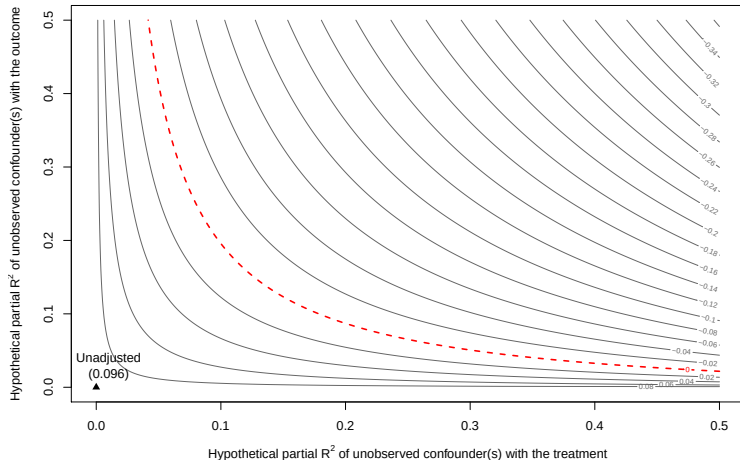
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- **VRF**: had we included Z , residual Y -variance would shrink.
- **VIF**: had we included Z , its correlation with D would inflate the SE.

First, good to have this in your head

Figure: Contour plot in the partial R^2 parameterization



Two simple statistics for routine reporting

Robustness Value (RV): the level of (equal) partial- R^2 association with D and Y that would eliminate the effect.

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Both come from t_D and df alone — you can compute them from a published regression table.

Augmented regression table: Darfur

Outcome: *Peace Index*

Treatment:	Est.	SE	t	$R^2_{Y \sim D X}$	RV	$RV_{\alpha=0.05}$
<i>Directly Harmed</i>	0.097	0.023	4.18	2.2%	13.9%	7.6%
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Somebody tell us:

- what does this RV mean?
- what does this $RV_{\alpha=0.05}$ mean?
- what does this $R^2_{Y \sim D|X}$ mean?

Bounding by comparison to observed covariates

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For an observed covariate X_j , define:

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“ Z is at most k_D times as predictive of D , and k_Y times as predictive of Y , as X_j is.” Given $\{X_j, k_D, k_Y\}$, this *bounds* both $R_{Y \sim Z | D, \mathbf{x}}^2$ and $R_{D \sim Z | \mathbf{x}}^2$, and hence the worst-case bias. Conservative for multiple, non-linear confounders.

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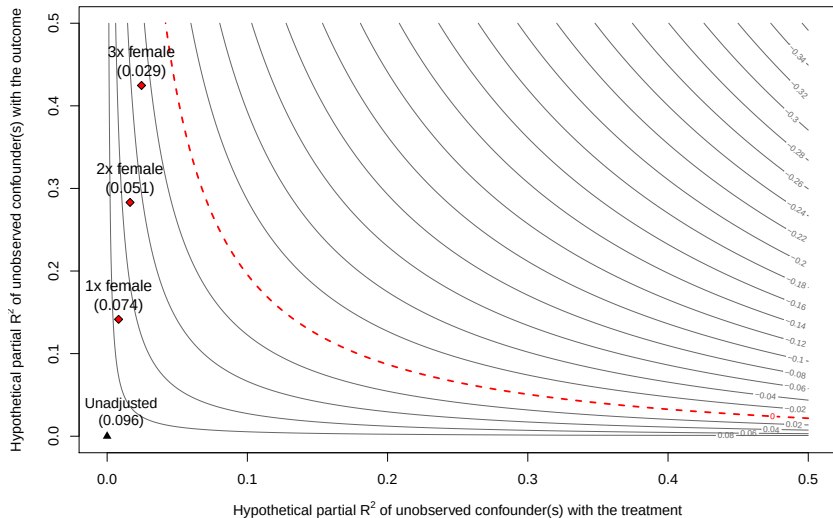
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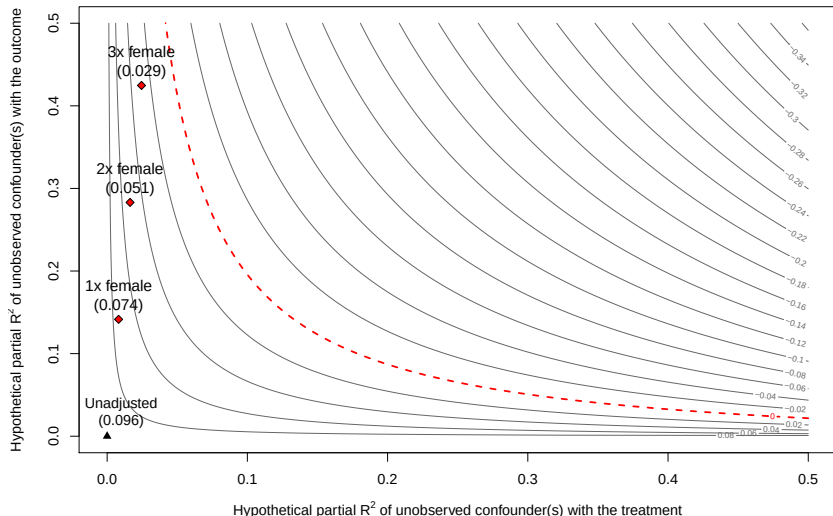
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Example: within Darfur villages, hard to imagine targeting on any basis stronger than *female*. So use $X_j = \text{female}$, and vary k .

Contour plot with bounds: Darfur



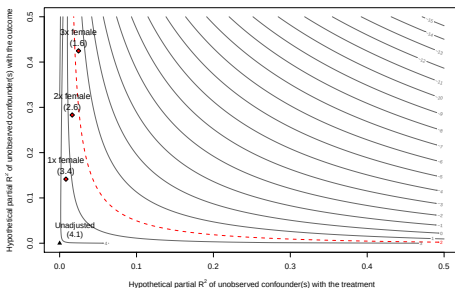
Contour plot with bounds: Darfur



Point estimate robust to a confounder 1 $\times$, 2 $\times$, or even 3 $\times$ as strong as *female*.

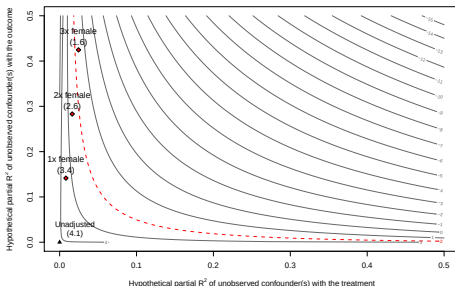
Inference and extreme scenarios

t -statistic contour



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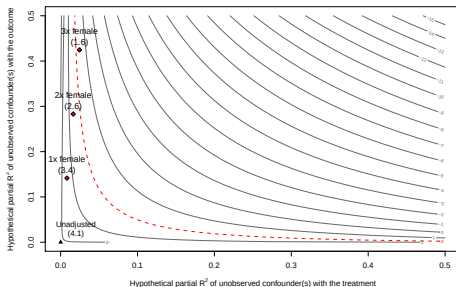
***t*-statistic contour**



t stays significant against a confounder up to
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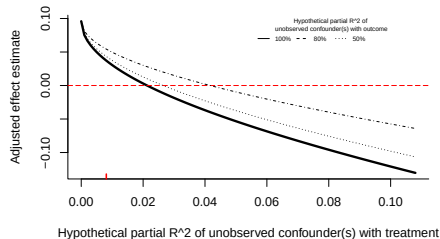
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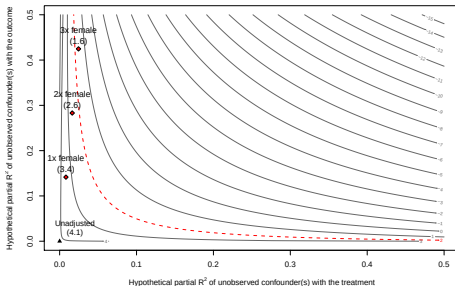
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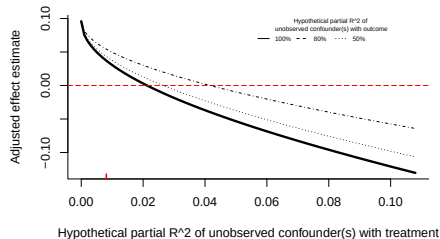
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Extreme scenario



Assume confounding explains *all* residual outcome variance ($R^2_{Y \sim Z | \mathbf{X}, D} = 1$). *Z* still needs to explain $\geq 2.2\%$ of *D* to zero the effect. You pay a real price when you can't bound.

How to argue it (and what readers should demand)

Defending a result. Sometimes you can argue from design and context, e.g. for Darfur:

*“... aircraft dropped makeshift and unguided bombs over villages, and militia raided without concern for who they would attack — the only known major exception, due to sexual assaults, was targeting gender, which is also one of the most visually apparent characteristics. **A confounder twice as strong as female would be surprising.**”*

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The reframe. This is never “how much confounding is there” — it’s “how much would it take to overturn the conclusion.” No magic RV threshold; the point is to refine the debate, not hide behind a number. Whether the answer is reassuring or unsettling, it’s the honest conversation.

Resources

Papers (all linked from chadhazlett.com/publications):

- Cinelli & Hazlett (2020). *Making sense of sensitivity: Extending omitted variable bias*. JRSS B. — the framework
- Hazlett & Parente (2023). *From “Is it unconfounded?” to “How much confounding would it take?”* JoP. — applied / how-to-argue companion
- Cinelli, Ferwerda & Hazlett (2024). *sensemakr: Sensitivity Analysis Tools for OLS in R and Stata*. Observational Studies. — software paper
- *Wildfire Exposure Increases Pro-Climate Political Behaviors* — worked applied example

Software & tutorials:

- `sensemakr` R/Stata + worked vignettes: carloscinelli.com/sensemakr
- Carlos Cinelli's site (talks, tutorials, videos): carloscinelli.com